Corey Scher

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Education

Ph.D. (*2025*) **Earth and Environmental Sciences.** CUNY Graduate Center.
Dissertation: "InSAR Monitoring of Landscape Changes in War and Conflict Settings with Case Studies in Ukraine, Gaza, and the Jordan Valley."

M.S. (*2019*) **Geology.** City College of New York.
Thesis: "Snowmelt Detection on Alpine Glaciers using Synthetic Aperture Radar Time Series."

B.A. (*2017*) **Geology.** University of California, Berkeley.

Professional Experience

2025– **Postdoctoral Researcher**, *Oregon State University.*
Damage mapping in conflict and disaster settings with open-source Earth observation data.

2024– **Geographic Information Systems (GIS) Innovation Advisor**, *United Nations Office for Project Services (UNOPS).*
Development of GIS capabilities within the UNOPS Peace and Security Cluster.

2018–2022 **Graduate Research Fellow**, *City College of New York.*
Project: Mapping seasonal glacier melt across the Hindu Kush Himalaya with time-series synthetic aperture radar (SAR).

2018–2020 **Graduate Research Fellow**, *Advanced Science Research Center.*
Project: Coastal Hypoxia Analysis and Risk Tracking (CHART) via remote sensing and process-based modeling in South and Southeast Asia.

2020 **Summer Intern**, *NASA Jet Propulsion Laboratory.*
Project: Advancing synergies between ECOSTRESS and SAR for structure/thermal regimes.

2017–2018 **Graduate Research Fellow**, *City College of New York.*
Project: NASA Pandora spectroscopy of atmospheric composition and Earth–atmosphere interactions.

2017–2019 **Research Affiliate**, *University of San Francisco.*
Project: Remote monitoring of groundwater overdraft with GRACE and InSAR.

2016–2017 **Undergraduate Research Apprentice**, *University of California, Berkeley.*

Project: Investigating hydraulic-scaling laws on river channel geometry using multi-frequency LiDAR.

Teaching

2017–2024 **Lab Instructor**, Earth Systems Science. City College of New York, Dept. of Earth and Atmospheric Sciences.

2018 **Lab Instructor**, Systems Analysis of the Earth. City College of New York, Dept. of Earth and Atmospheric Sciences.

Workshop Instruction

2024 **NASA and USAID**. Three-day workshop on urban hazard and vulnerability assessment with Earth observation data, San Salvador, El Salvador.

2022 **NASA Applied Remote Sensing Training (ARSET)**. "Monitoring Urban Damage with Interferometric Synthetic Aperture Radar."

Awards and Fellowships

2026 **Open Science Recognition Prize.** American Geophysical Union award to the Decentralized Damage Mapping Group for sharing satellite-derived conflict and disaster damage data with humanitarian responders and journalists.

2025 **Earth Observation (EO) Excellence Team Award.** European Space Agency award to the Decentralized Damage Mapping Group for advancing EO-based conflict damage mapping.

2020 **NASA Summer Maximizing Student Potential in STEM Program**. Supported a summer research internship at the NASA Jet Propulsion Laboratory.

2019 **American Geophysical Union (AGU) Travel Grant Recipient.** Conference: "Quest for Sustainability of Heavily Stressed Aquifers," Valencia, Spain.

Grants and Funding

2026–2029 **Co-Investigator.** "Uncertainty-Aware NISAR Damage Mapping for Disaster Response." NASA Earth Action. PI: Jamon Van Den Hoek.

Publications

1. Sticher, Valerie, Corinne Bara, Jenniina Kotajoki, Lars Bromley, Thierry Crevoisier, Emanuele Dalsasso, Olivier Dietrich, Karsten Donnay, Filip Dorm, Manuel Fiol, Xi Li, Mikael Hiberg Naghizadeh, **Corey Scher**, Jamon Van Den Hoek, Konrad Schindler, Devis Tuia, and Jan Dirk Wegner. "Advancing Conflict Research and Response Through Satellite-Derived Data." *Nature* (2026). doi:10.1038/s41586-026-11004-6.

2. Witmer, Frank D. W., **Corey Scher**, Jamon Van Den Hoek, and John O'Loughlin. "Measuring and Accounting for War Destruction Using Human-Reported and Satellite-Derived Conflict Data: Ukraine, 2022–2023." *Annals of the American Association of Geographers* (2026): 1–14.

3. Rinaldo, Daniele, Rami Alazzeh, Jean-Louis Arcand, **Corey Scher**, and Jamon Van Den Hoek. "The Destruction of Gaza: Satellite Measurements of the Economic Cost of War." *PNAS Nexus* 5, no. 5 (2026): pgag124.
4. Mills, David, Yara Asi, Dennis Kunichoff, Erica Nelson, Juliana Nasser, Shannon Barkley, Ryan Arias Delafosse, **Corey Scher**, et al. "Hospital Functionality in the Gaza Strip: Analysis of the World Health Organization Health Resources and Services Availability Monitoring System Database, 7 Oct 2023 to 31 Oct 2024." *Conflict and Health* 20, no. 1 (2026): 35.
5. Ulbricht, Bailey, Allen S. Weiner, Jamon Van Den Hoek, and **Corey Scher**. "There Is Nothing Left: Jus Ad Bellum Proportionality and Israel's War Against Hamas in Gaza." *Berkeley Journal of International Law* (2026).
6. **Corey Scher** and Jamon Van Den Hoek. "Nationwide Conflict Damage Mapping with Interferometric Synthetic Aperture Radar: A Study of the 2022 Russia–Ukraine Conflict." *Science of Remote Sensing* (2025): 100217.
7. **Scher, Corey.** "InSAR Monitoring of Landscape Changes in War and Conflict Settings with Case Studies in Ukraine, Gaza, and the Jordan Valley." Ph.D. dissertation, *City University of New York Graduate Center*, 2025.
8. Asi, Yara, David Mills, P. Gregg Greenough, Dennis Kunichoff, Saira Khan, Jamon Van Den Hoek, **Corey Scher**, et al. "'Nowhere and No One Is Safe': Spatial Analysis of Damage to Critical Civilian Infrastructure in the Gaza Strip During the First Phase of the Israeli Military Campaign, 7 October to 22 November 2023." *Conflict and Health* 18, no. 1 (2024): 24.
9. **Corey Scher**, Nick Steiner, and Kyle McDonald. "Mapping Seasonal Glacier Melt Across the Hindu Kush Himalaya with Time Series SAR." *The Cryosphere* 15, no. 9 (2021): 4465–4482.

Preprints and Working Papers

10. **Scher, Corey** and Jamon Van Den Hoek. "Active InSAR Monitoring of Building Damage in Gaza During the Israel–Hamas War." *Preprint*, arXiv:2506.14730, 2025.

Invited Talks and Guest Lectures

Aug 2026 **Invited panelist**, "Towards an AI Ready Geospatial Ecosystem," side event of the *UN Committee of Experts on Global Geospatial Information Management (UN-GGIM)* in advance of the *Third UN World Geospatial Information Congress*, United Nations Headquarters, New York, NY.

Jul 2026 **Invited talk**, satellite-based damage mapping of the 2026 Venezuela earthquakes, *GIS Sub-Group of the Global Information Management Working Group (IMWG)*, hosted by the *UN Office for the Coordination of Humanitarian Affairs (OCHA)*, virtual.

Nov 2025 **Guest Lecture**, "Tracking progressive war damage with satellite radar," *School of Foreign Diploy, Georgetown University*, Washington, D.C.

Oct 2025 **Invited talk**, "Monitoring Landscape Changes in Armed Conflict Zones," *Bloomberg News*, New York, NY.

Sep 2025 **Invited talk**, "AI for Global Progress – An Innovation Showcase," hosted by *Google Public Sector* during the *United Nations General Assembly High-Level Week*, in affiliation with *UNOPS*, New York, NY.

Jun 2025 **Student Speaker**, CUNY Graduate Center Commencement Ceremony (link)

May 2025 **Invited talk**, "Mapping War Damage from Space," *Nerd Nite*, CUNY Graduate Center.

Apr 2025 **Invited talk**, "Landscape Changes in Armed Conflict Zones and Disaster Settings with Satellite Radar Data," *Sixth Annual Image Intelligence Workshop*, *United Nations Global Service Centre*.

Apr 2025 **Guest lecture**, "Mapping Wartime Building Damage with Interferometric Synthetic Aperture Radar," *University of Wisconsin–Madison*.

Apr 2025 **Invited panelist**, "Environmental Impacts and Implications of War in Ukraine," *Ukrainian Institute of America* with *Ukrainian Jersey City*, New York, NY.

Mar 2025 **Guest lecture**, "Mapping Wartime Building Damage with Interferometric Synthetic Aperture Radar," Dept. of Electrical Engineering, *The Cooper Union*.

Feb 2025 **Invited panelist**, "Digital Landscapes," *Harvard Ukrainian Research Institute*, Harvard University.

Dec 2024 **Invited talk**, "Damage Mapping," *Humanitarian Intelligence Unit*, U.S. Department of State.

Oct 2024 **Invited talk**, "Persistent Monitoring of Damage to Built-up Areas in Gaza During the First Year of the 2023 Israel–Hamas War," *Workshop on GIS and LIS Curriculum*, Pratt Institute, New York, NY.

Nov 2023 **Invited talk**, "Mapping Damage in Gaza Using Sentinel-1 Coherence," *Financial Times Weekly Graphics Team Meeting*.

Nov 2023 **Invited talk**, "Brown Bag Lunch Conversation," *New York Times Graphics Team Meeting*.

Nov 2023 **Invited talk**, "Using Remote Sensing to Assess Damage in Gaza," *Foreign, Commonwealth, and Development Office of the United Kingdom*.

Media and Public Attention

Research and data products featured, cited, or discussed in more than 500 international media outlets; also included in reports from international humanitarian organizations and cited in international legal proceedings. Full list here; more at whoiscorey.com.

Selected Media

Oct 2025 *ABC*: "Satellite Images Show Extent of Destruction in Gaza After Two Years of War." (link)

Oct 2025 *BBC*: "Gaza War in Maps and Satellite Images." (link)

Jul 2025 *Haaretz*: "500 Missiles, 200 Interceptors, $1.5 Billion: Numbers Behind Iran's Attacks on Israel." (link)

Mar 2025 *Bloomberg*: "Satellite Imagery Shows Gaza's Destruction and Resilience With War Unresolved." (link)

Mar 2025 *Financial Times*: "Lebanon's Displaced Return to Find 'Scorched Earth' After Israeli Offensive." (link)

Jan 2025 *The New York Times*: "Mapping the Damage in Altadena and Pacific Palisades." (link)

Oct 2024 *The New York Times*: "Gaza in Ruins After a Year of War." (link)

Aug 2024 *Financial Times*: "'We Thought We Lived in Tuscany': Hizbollah Strikes Ravage Israel's North." (link)

Jun 2024 *The New York Times*: "What Ukraine Has Lost." (link)

Jun 2023 *The Washington Post*: "Rural Areas Sacrificed for Xi Jinping's New City, Satellite Imagery Shows." (link)

Selected Interviews and Appearances

Oct 2025 *BBC Radio 4, The World at One*: "The Challenge of Rebuilding Gaza." (link)

Aug 2025 *Der Standard*: "War cartographers Scher and Van Den Hoek: Destruction in Gaza worse than in Dresden 1945." (link)

Jul 2025 *BBC Radio 4, The World This Weekend*: "A Picture of Life in Gaza." (link)

Feb 2025 *The Thought Project Podcast, CUNY Graduate Center*: "Mapping the Wreckage of War and Wildfires." (link)

Jan 2024 *Marketplace*: "How Satellite Radar Helps Scientists Map the Destruction in Gaza." (link)

Jan 2024 *Bloomberg News*: "MapLab: Mapping Gaza's Destruction." (link)

Dec 2023 *Scientific American*: "Inside the Satellite Tech Revealing Gaza's Destruction." (link)

Nov 2023 *Democracy Now!*: "Gaza in Ruins: Satellite Imagery Researchers Say Israel Has Destroyed or Damaged 56,000 Buildings." (link)

Selected Reports and Legal Proceedings

Oct 2024 *United Nations General Assembly*: "Economic costs of the Israeli occupation for the Palestinian people: the economic impact of the Israeli military operation in Gaza from October 2023 to May 2024"
Incorporates satellite-based building damage data and analysis from Scher & Van Den Hoek. (link)

Jan 2024 *United Nations Conference on Trade and Development (UNCTAD)*: "Preliminary assessment of the economic impact of the destruction in Gaza and prospects for economic recovery: UNCTAD rapid assessment."

Incorporates satellite-based building damage data and analysis from Scher & Van Den Hoek. (link)

Dec 2023 *International Court of Justice*: "Application of the Convention on the Prevention and Punishment of the Crime of Genocide in the Gaza Strip (South Africa v. Israel)"
References satellite-based building damage data and analysis from Scher & Van Den Hoek. (link)

Conference Presentations

1. Barthelme, Philipp, **Corey Scher**, Myscon Truong, He Yin, and Jamon Van Den Hoek. "InSAR+: Exploring the Utility of Complementary Data Sources for Mapping Conflict Damage Using InSAR Coherence Time Series." Presented at the *EGU General Assembly 2026*, Vienna, Austria, May 2026. EGU26-19597, doi:10.5194/egusphere-egu26-19597.
2. Van Den Hoek, Jamon, and **Corey Scher**. "Nationwide Mapping of Damage to Human Settlements Across Ukraine Using Sentinel-1 InSAR Coherent Change Detection." Paper presented at the *Global Land Program Open Science Meeting*, Online, November 5, 2024.
3. **Scher, Corey**. "The Science and Practice of War Damage Impact Assessment Using Satellite Radar." Application Lightning Talk presented at *SatSummit 2024*, Washington, D.C., May 17, 2024.
4. **Scher, Corey**, and Jamon Van Den Hoek. "Nationwide Mapping of Damage to Human Settlements Across Ukraine Using Sentinel-1 InSAR Coherence Change Detection." Talk presented at the *AGU Fall Meeting*, San Francisco, CA, December 2023
5. **Scher, Corey**, and Jamon Van Den Hoek. "Decentralized, Nation-Wide, High-Frequency War Damage Mapping Using InSAR Time Series Data." Poster presented at the *AGU Fall Meeting*, Chicago, IL, December 2022.
6. **Scher, Corey**, Kyle C. McDonald, Charles J. Vorosmarty, and David Saah. "Monitoring Sustainability of Shared Groundwater Resources Using Satellite Measurement of Transboundary Aquifer Compaction." Talk presented at the *Chapman Conference on the Quest for Sustainability of Heavily Stressed Aquifers at Regional to Global Scales*, American Geophysical Union, Valencia, Spain, October 2019.
7. **Scher, Corey**, and David Saah. "Extent and Characteristics of Damage from Wildfires Caused by Incendiary Kites During Protests of the Gaza–Israel Barrier Fence (March 2018 to Present)." Poster presented at the *AGU Fall Meeting*, San Francisco, CA, December 2018.
8. **Scher, Corey**, and David Saah. "Remote Monitoring of Groundwater Overdraft Using GRACE and InSAR." Poster presented at the *AGU Fall Meeting*, San Francisco, CA, December 2017.
9. **Scher, Corey**, Christopher Tennant, Laurel Larsen, and Dino G. Bellugi. "Do Rivers Really Obey Power-Laws?: Using continuous high resolution measurements to define bankfull channel and evaluate downstream hydraulic-scaling over large changes in drainage area." Poster presented at the *AGU Fall Meeting*, San Francisco, CA, December 2016.

Service

Program Service

2021–2025 **Executive Committee Student Member.** Dept. of Earth and Environmental Sciences, CUNY Graduate Center.

2019–2021 **Admissions Committee Student Member.** Dept. of Earth and Environmental Sciences, CUNY Graduate Center.

Professional Service

Conference Sessions

Dec 2025 **Co-Convener**, "Damage Assessment Across Scales and Systems: Environmental and Societal Impacts of Conflict and Disaster," *AGU 2025 Fall Meeting.*

Dec 2024 **Primary Convener**, "Tracking the Footprint of War: Earth and Environmental Science Perspectives on Monitoring Conflict Impacts," *AGU 2024 Fall Meeting.*

Academic Journal Editing

2025 **Guest Editor**, *Environmental Research Letters* special issue: "Initial and Enduring Environmental Consequences of Armed Conflict" (forthcoming).

Academic Journal Reviewing
Science Advances — Geophysical Research Letters — Water Resources Research — Science of Remote Sensing

Professional Affiliations

- American Geophysical Union

Software Development, Programming, & Field Experience

- Expert in Python, JavaScript, and shell scripting for geospatial data science and remote-sensing time-series analysis
- Expert in InSAR; engineering of scalable cloud-based pipelines using ASF HyP3, Google Cloud, and Google Earth Engine
- Data visualization with Python, Adobe Creative Suite, and GIS platforms
- Formal training in field geologic mapping and hydrogeology
- Field-based atmospheric remote sensing; water-optics sampling and sensor deployment (NASA Pandora network)
- In situ experiment on vegetation-induced sedimentation, Wax Lake Delta, Louisiana (PI: Laurel Larsen)